

# PANDAS VISUAL + COMBINE + RESHAPE API

CONCEPT NAME -> syntax only | packed combined sheet

## SEABORN BIVARIATE

```
sns.scatterplot(x='total_bill', y='tip',
               data=tips)
sns.regplot(x='total_bill', y='tip', data=tips)
sns.regplot(x='total_bill', y='tip', data=tips,
            fit_reg=False)
sns.jointplot(x='total_bill', y='tip',
             data=tips)
sns.jointplot(x='total_bill', y='tip',
             data=tips,
             kind='hex')
sns.jointplot(x='total_bill', y='tip',
             data=tips,
             kind='kde')
sns.barplot(x='time', y='total_bill',
            data=tips)
sns.barplot(x='time', y='total_bill',
            data=tips,
            estimator='std')
sns.boxplot(x='time', y='total_bill',
            data=tips)
sns.violinplot(x='time', y='total_bill',
              data=tips)
```

## RESHAPING PAIRS

```
wide -> pd.melt(...)
long -> df.pivot_table(...)
split -> s.str.split(...)
attach -> pd.concat(..., axis=1)
dedupe -> df.drop_duplicates()
link -> df.merge(...)
```

## CONCAT COLUMNS

```
pd.concat([df1, df2, df3], axis=1)
col_concat['A']
col_concat['new_col'] = ['n1','n2','n3','n4']
pd.concat([df1, df2], axis=1,
          ignore_index=True)
```

## SUFFIXES

```
left.merge(right, on='key', suffixes=('_left',
                                       '_right'))
left.merge(right, left_on='a', right_on='b',
           suffixes=('_x', '_y'))
```

## MERGE BASIC

```
left.merge(right, on='shared_col')
left.merge(right, left_on='name',
           right_on='site')
```

## IMPORTS

```
import pandas as pd
import glob
```

## FACETS

```
sns.lmplot(x='x', y='y', data=anscombe,
           fit_reg=False,
           col='dataset', col_wrap=2)
sns.lmplot(x='total_bill', y='tip', data=tips,
           fit_reg=False, hue='sex', col='day')
facet = sns.FacetGrid(tips, col='time')
facet.map(sns.histplot, 'total_bill')
facet = sns.FacetGrid(tips, col='day',
                     hue='sex')
facet.map(plt.scatter, 'total_bill', 'tip')
facet.add_legend()
facet = sns.FacetGrid(tips, col='time',
                     row='smoker',
                     hue='sex')
sns.catplot(x='day', y='total_bill', hue='sex',
            data=tips, row='smoker', col='time',
            kind='violin')
```

## MULTIPLE FILES

```
glob.glob('data/*.csv')
files = glob.glob('data/*.csv')
pd.read_csv(f)
dfs = [pd.read_csv(f) for f in files]
pd.concat(dfs)
pd.concat([pd.read_csv(f) for f in files],
          ignore_index=True)
```

## STYLES

```
sns.set_style('whitegrid')
sns.set_style('darkgrid')
sns.set_style('dark')
sns.set_style('white')
sns.set_style('ticks')
```

## CONCAT LABELS

```
pd.concat([df1, df2, df3], join='outer')
pd.concat([df1, df2, df3], join='inner')
pd.concat([df1, df3], join='inner')
pd.concat([df1, df2, df3], axis=1)
pd.concat([df1, df3], axis=1, join='inner')
```

## MELT BASIC

```
pd.melt(pew, id_vars='religion')
pd.melt(pew, id_vars='religion',
       var_name='income',
       value_name='count')
```

## ONE-TO-ONE

```
visited_subset = visited.loc[[0, 2, 6], :]
site.merge(visited_subset, left_on='name',
           right_on='site')
```

## SPLIT STRINGS

```
variable_split =
    ebola_long['variable'].str.split('_')
ebola_long['status'] =
    variable_split.str.get(0)
ebola_long['country'] =
    variable_split.str.get(1)
variable_split =
    ebola_long['variable'].str.split('_',
    expand=True)
variable_split.columns = ['status','country']
pd.concat([ebola_long, variable_split], axis=1)
ebola_long['status'], ebola_long['country'] =
    zip(*ebola_long['variable'].str.split('_'))
```

## PIVOT TABLE

```
weather_melt.pivot_table(index=['id','year'],
                          'day', columns='element', values='temp')
weather_melt.pivot_table(..., aggfunc='mean')
weather_tidy =
    weather_melt.pivot_table(...).reset_index()
df.pivot_table(index='id', columns='var',
               values='val').reset_index()
```

## SEABORN UNIVARIATE

```
sns.histplot(tips['total_bill'])
sns.histplot(tips['total_bill'], kde=True)
sns.kdeplot(tips['total_bill'])
sns.rugplot(tips['total_bill'])
sns.countplot(x='day', data=tips)
```

## MATPLOTLIB PLOTS

```
ax.hist(tips['total_bill'], bins=10)
ax.scatter(tips['total_bill'], tips['tip'])
ax.boxplot(..., labels=[])
ax.scatter(x=..., y=..., s=..., c=...,
          alpha=0.5)
```

## INDEX / LABEL CHECKS

```
row_concat.loc[3]
row_concat.loc[0]
df1.columns = ['A','B','C','D']
df1.index = [0,1,2,3]
```

## MELT WEATHER

```
weather_melt = pd.melt(weather,
                      id_vars=['id','year',
                              'month','element'], var_name='day',
                      value_name='temp')
```

## DO NOT USE

```
sns.distplot(...)
```

## NORMALIZE

```
billboard_songs =
    billboard_long[['year','artist',
                    'track','time']].drop_duplicates()
billboard_songs['id'] =
    range(len(billboard_songs))
billboard_ratings =
    billboard_long.merge(billboard_songs,
    on=['year',
        'artist','track','time'])
billboard_ratings = billboard_ratings[['id',
    'date.entered','week','rating']]
```

## AX METHODS

```
ax.plot(x, y)
ax.plot(x, y, 'o')
ax.set_title('Title')
ax.set_xlabel('x')
ax.set_ylabel('y')
fig.suptitle('Title')
fig.tight_layout()
plt.show()
```

## MANY-TO-MANY

```
visited.merge(survey, left_on='ident',
              right_on='taken')
site.merge(visited, left_on='name',
           right_on='site').merge(survey,
                                left_on='ident',
                                right_on='taken')
```

## CONCAT ROWS

```
pd.concat([df1, df2, df3])
pd.concat([df1, df2, df3], ignore_index=True)
pd.concat([df1, new_row_df])
pd.concat([df1, pd.DataFrame([data_dict])],
          ignore_index=True)
```

## MANY-TO-ONE

```
site.merge(visited, left_on='name',
           right_on='site')
survey.merge(person, left_on='person',
             right_on='ident')
```

## IMPORTS

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

## DO NOT USE

```
df.append(df2)
df1.append(data_dict, ignore_index=True)
```

## PANDAS PLOT

```
df['col'].plot()
df['col'].plot.hist()
df.plot.scatter(x='a', y='b')
df.plot.box()
df.plot.bar()
df.plot.line()
df.plot.area()
df.plot.density()
df.plot.hexbin(x='a', y='b')
```

## MULTIVARIATE

```
sns.violinplot(x='time', y='total_bill',
              hue='sex',
              data=tips, split=True)
sns.lmplot(x='total_bill', y='tip', data=tips,
          hue='sex', fit_reg=False)
sns.lmplot(..., markers=['o','x'],
          scatter_kws={'s':
              tips['size'] * 10})
```

## PAIRWISE

```
sns.pairplot(tips)
sns.pairplot(tips, hue='sex')
grid = sns.PairGrid(tips)
grid.map_upper(sns.scatterplot)
grid.map_lower(sns.kdeplot)
grid.map_diag(sns.histplot)
```

## FIGURE / AXES

```
fig = plt.figure()
ax = fig.add_subplot(1, 1, 1)
fig, ax = plt.subplots()
fig, axes = plt.subplots(2, 2, figsize=(8, 8))
axes[0, 0].plot(...)
```

## MERGE HOW

```
left.merge(right, on='key', how='inner')
left.merge(right, on='key', how='left')
left.merge(right, on='key', how='right')
left.merge(right, on='key', how='outer')
```

## MELT MULTI ID

```
pd.melt(billboard,
       id_vars=['year','artist','track',
               'time','date.entered'], var_name='week',
       value_name='rating')
```

## LOAD DATA

```
anscombe = sns.load_dataset('anscombe')
tips = sns.load_dataset('tips')
```

## IMPORT

```
import pandas as pd
```